

## Architectural Justification — Warehouse Inventory & Pallet Location Tracker

*We chose Layered Architecture for the Warehouse Inventory & Pallet Location Tracker System.*

### Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (Operator Scan Interface, Supervisor Dashboard), a Business Layer (Pallet Placement Manager, RFID/Barcode Scan Adapter, Weight Limit Validator, Movement Logging & Capacity Alert Service), and a Data Layer (Warehouse Database). Each layer depends only on the layer directly beneath it, and all cross-cutting rules — weight-limit validation, occupancy checks, and movement logging — are enforced centrally in the Business Layer rather than duplicated across client devices.

### Reason 1: Centralized, Consistent Business Rule Enforcement

FR-001 and FR-004 require that every pallet placement be validated against rack capacity and shelf occupancy before it is accepted, and FR-005 requires a 90% capacity alert. A layered design lets the Pallet Placement Manager and Weight Limit Validator apply these rules once, in one place, for every scan event coming from any operator handheld or kiosk — eliminating the risk of inconsistent validation that a fully distributed (microservices) design would introduce across independently deployed services.

### Reason 2: Clean Separation Between Hardware, Logic, and Storage

The scenario mixes physical scanning hardware, business workflow, and persistent tracking data. Isolating the RFID/Barcode Scan Adapter in the Business Layer behind a clear Scan Event Interface means the scanning hardware can be replaced or upgraded (e.g., new RFID readers) without touching placement logic or the database schema, directly matching the component modularity and replaceability principles required by this lab.

### Security Advantage

Because the Presentation Layer never talks to the Warehouse Database directly, both operator and supervisor clients must pass through the Business Layer's Pallet Placement Manager and Movement Logging service. This lets a single point enforce authentication, role-based access (e.g., only Logistics Supervisors can view rack-wide capacity alerts), and input validation, shrinking the attack surface compared to giving every client direct data-layer access.

### Performance Benefit

NFR-001 requires SKU lookups across 100,000 bins in under 100 ms, and NFR-002 requires scan data synchronized within 2 seconds. Keeping the Data Layer as a single, well-indexed component that only the Business Layer queries avoids the network hops and eventual-consistency delays of a distributed data store, making sub-100ms lookups and near-real-time sync easier to guarantee and to benchmark under NFR-002's 99.5% uptime target.